

COURSE TITLE : SPECIALITY POLYMERS
COURSE CODE : 6096
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER: 60/6
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	High temperature and fire resistant polymers	15
2	Polymers in electrical and electronics applications.	20
3	Ionic polymers & Nano polymer composites.	15
4	Polymer concrete and Polymer binders	10
TOTAL		60

COURSE OUTCOME

Sl.	Sub	Student will be able to
1.	1	To understand and identify the high temperature and fire resistant polymers.
2	2	To identify the applications of polymers in electrical and electronics fields.
3	3	To understand the preparation and properties of ionic & nano polymers.
4	4	To know the applications of polymer concrete and polymer binders in various areas.

SPECIFIC OUTCOME

MODULE I

1.1.0 High temperature and fire resistant Polymers

- 1.1.1 Factors affecting the thermal stability and fire resistance of polymers.
- 1.1.2 Explain the methods for improving the performance of polymers for high temperature use
- 1.1.3 Heat and fire resistant polymers
 - 1.1.3.1 Fluoro polymers
 - 1.1.3.2 Aromatic Polymers
 - 1.1.3.3 Poly phenylene Sulphide
 - 1.1.3.4 Poly sulphones
 - 1.1.3.5 Polyesters

- 1.1.3.6 Polyamides
- 1.1.3.7 Poly ketones

MODULE II

2.1.0 Polymers used in electrical and electronics applications

- 2.1.1 Define Conducting polymers
- 2.1.2 Describe the electrical and electronic properties of polymers
- 2.1.3 Explain the conducting mechanism of polymers
- 2.1.4 Understand the structure, preparation, properties and application of conducting polymers like
 - 2.1.4.1 Polyacetylene
 - 2.1.4.2 Polyparaphenylene
 - 2.1.4.3 Polypyrrole
 - 2.1.4.4 Organometallic polymers
- 2.1.5 Explain about photo conducting polymer
- 2.1.6 Understand Polymers with piezo electric and pyro electric properties
- 2.1.7 Explain the photo resists for semiconductor application.
- 2.1.8 Explain briefly liquid crystalline polymers

MODULE III

3.1.0 Ionomers & Nano Polymers

- 3.1.1 Explain the production properties and application of ionic polymers
- 3.1.2 Understand the structure, preparation and properties of of ionomers based on
 - 3.1.2.1 Polystyrene
 - 3.1.2.2 Polyaromatic backbones
 - 3.1.2.3 PTFE
- 3.1.3 Explain the polyelectrolyte ion exchangers

3.2.0 Nano polymers

- 3.2.1 Nano materials & Structure
- 3.2.2 Nano material processing & applications

MODULE IV

4.1.0 Understand Polymer concrete and polymer binders

- 4.1.1 Explain about polymer concrete
- 4.1.2 Explain the production, properties of polymer impregnated concrete ,ultra high modulus fibres
- 4.1.3 List out any four polymers used in bio-medical application, Explain their production and properties.
- 4.1.4 Explain the polymers used as rocket propellants. Give examples.
- 4.1.5 Explain the production of optical fibres used in telecommunication cables
- 4.1.6 Method of wave signal communication optical in fibre

CONTENT DETAILS

MODULE-I

High temperature Polymers

High temperature and fire resistant polymers improving low performance polymers for high temperature use – polymers, for low fire hazards – polymers for high temperature resistance – Fluoropolymers. Aromatic polymers, polyphenylene sulphide, polysulphones, polyesters, polyamides, polyketones, Heterocyclic polymers.

MODULE II

Polymers with electrical and electronic properties Conducting polymers, conducting mechanisms, polyacetylene, polyparaphenylene polypyrrole, organometallic polymers, photo conducting polymers, polymers in non-linear optics, polymers with piezoelectric pyroelectric and pyroelectric properties, photoresists for semi conductor fabrication – liquid crystalline polymers.

MODULE III

Ionic Polymers, synthesis, physical properties and applications, Ion-exchange, Hydrophilicity, Ionomers based on polyethylene, elastomeric ionomers. Ionomers based on polystyrene, ionomers based on PTFE, ionomers with polyaromatic backbones, polyelectrolytes for ion exchange, polyelectrolytes based on carboxylates, nano materials-structure-processing and application

MODULE IV

Polymer concrete, polymer impregnated concrete ultra high modulus fibres, polymers for biomedical applications, polymeric binders for rocket propellants, polymer supported reagents. Optical fibre telecommunication cables.

REFERENCE BOOKS:

1. H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & Sons, New York, 1989.
2. Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985.
3. R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998.
4. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel Dekker, New York, 1993
5. NANO COMPOSITES –